



A NEAR-INFRARED PHOTODETECTOR TUNED BY QUANTUM DOT SIZE

MATERIALS PROCESSING & OPTOELECTRONICS

TOOLS & METHODS:

Hot-injection colloidal synthesis · layer-by-layer spin coating with TBAI ligand exchange · HCl/Zn wet etch of ITO · graphene wet transfer · Hitachi UH4150 UV-Vis-NIR spectrophotometry (2001 points per scan, 0.5 nm interval) · Keithley 2400 SourceMeter I-V and four-point sheet-resistance extraction · Keysight 33500B gated 635 nm laser photoresponse · rise/fall, EQE and responsivity analysis.

EXECUTIVE SUMMARY:

Near-infrared detection matters for optical communications, night vision, LiDAR and remote sensing, but the market forces a choice: inexpensive silicon photodiodes stop near 1100 nm, while InGaAs and germanium devices reach 1700 to 1800 nm at roughly ten times the price. Lead-sulfide quantum dots offer a way around that trade-off, because their band gap, and so the wavelength they absorb, is set by particle size rather than by material choice.

Three PbS quantum-dot sizes were synthesized, their band gaps confirmed by UV-Vis spectrophotometry, and a stacked detector built in which the smallest dots sit in front so each layer absorbs the photons it is tuned for and passes the rest through. The architecture achieved 500 to 1500 nm coverage and cut dark current 44-fold against a previous gold-on-silicon build. Photocurrent and quantum efficiency remain below commercial parts, and a full characterization suite traces every shortfall to a specific, correctable cause.

KEY RESULTS:

- **Three QD sizes** from oleic-acid ratios of 1:2, 1:8 and 1:24, with peaks at 805, 1188 and 1426 nm and band gaps of **1.54, 1.04 and 0.87 eV**.
- **Stacked device covers 500 to 1500 nm**; individual chips span 500 to 1100, 1100 to 1300 and 1300 to 1500 nm.
- **Dark current reduced 44 times** (415 to 9.3 μA) by replacing gold-on-silicon with transparent glass/ITO.
- **Spectral filtering confirmed electrically**: stacking halves the medium chip's photocurrent and cuts the large chip's by 80%.
- **Sheet resistance rises 15 to 40%** after TBAI ligand exchange, traced to graphene π -bond interactions.
- Photocurrent 136 nA, responsivity 114 $\mu\text{A/W}$, EQE 0.022% at 635 nm and 1.2 mW.

INTRODUCTION:

Quantum confinement makes the PbS band gap a function of nanocrystal diameter: smaller dots absorb shorter wavelengths, larger dots reach deeper into the infrared. The oleic-acid-to-lead ratio during hot-injection synthesis sets that diameter, so one low-cost chemistry of lead oxide, a sulfur precursor, oleic acid and octadecene yields a family of absorbers spanning the near infrared. Organic NIR photodetectors, by contrast, need four to six different polymers to cover the same range.

What the earlier build got wrong. A first-generation device stacked all three dot sizes on a single gold-coated silicon chip, two layers per size. Gold is an excellent contact: its conductivity of 4.11×10^7 S/cm against ITO's 1.3×10^4 S/cm let carriers move freely from the graphene channel into the electrode, and the device produced 9.8 μA of photocurrent with a rise time of 0.46 s. But the same conductivity produced 415 μA of dark current, the silicon substrate was opaque so no layer could be measured on its own, there was no guarantee carriers generated in the top layer ever reached the electrode, and gold sat badly with the cost target.

What changed. The second generation splits the stack into three separate transparent chips, each glass / ITO / graphene / PbS QD, carrying one dot size and characterized independently before assembly. The substrate is now transparent, so light passes through each chip to the next and every layer can be measured alone and in the stack. The trade is real and worth stating plainly: dark current fell 44-fold, but photocurrent fell with it, because ITO presents a small energy barrier at the graphene interface that gold did not.

TEST SETUP:

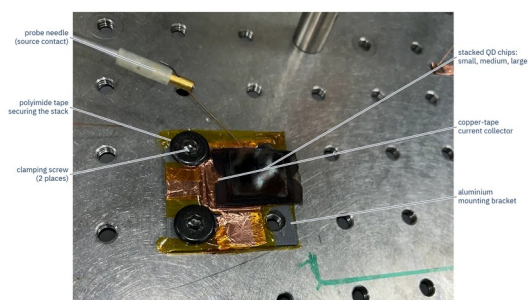


FIG 1: The assembled detector under test. Three transparent chips are stacked smallest-band-gap-last, clamped between an aluminium bracket and polyimide tape, with copper tape carrying current to the probe.

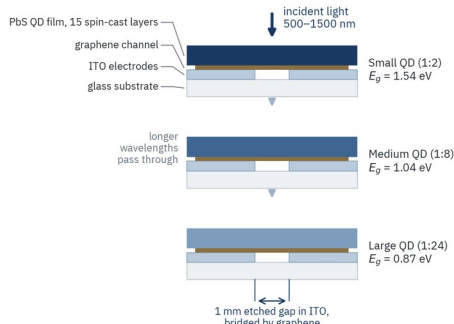


FIG 2: Layer structure. Each chip is glass / ITO / graphene / PbS QD, with the ITO wet-etched to leave a 1 mm gap bridged by transferred graphene.

Synthesis. Lead(II) oxide (0.94 g) in 25 mL octadecene with the target oleic-acid ratio was held at 90 °C for one hour, then a sulfur precursor was rapidly injected; the immediate change to black marked nucleation. Growth was quenched after 2 to 5 minutes at 82 °C and the dots purified by two to three toluene/acetone centrifugation cycles, dried under nitrogen, redispersed in toluene and syringe-filtered.

Device fabrication. ITO-coated glass in three sizes (17.6 × 14.7, 24.6 × 16.6 and 28.9 × 18.8 mm) was masked and wet-etched with 1 M HCl and zinc powder to open a 1 mm insulating gap, then graphene was wet-transferred across the gap to form the channel. Each chip received fifteen cycles of a three-step spin-coating recipe, 100 µL QD solution at 2000 rpm, tetrabutylammonium iodide (TBAI) ligand exchange and a methanol rinse at 4000 rpm, giving 45 QD layers across the assembled stack.

Characterization suite. Four measurements were run on every chip alone and in the stack. UV-Vis-NIR absorbance from 500 to 1500 nm at 0.5 nm resolution (2001 points per scan, noise below ± 0.002 Abs) gave the absorption peaks and, through $E_g = 1240 \text{ eV} \cdot \text{nm} / \lambda$, the band gaps. I-V sweeps from -1 to +1 V under ambient light and in the dark gave channel resistance, converted to sheet resistance through $R_s = (W/L)R = 5R$ for the 5.0 × 1.0 mm channel geometry. Gated photoresponse used a 635 nm, 1.2 mW laser switched by a 100 MHz square wave while a second SourceMeter logged current, giving

photocurrent $I_p = I_{on} - I_{off}$, and 10 to 90% rise and fall times. From those, external quantum efficiency and responsivity follow directly.

RESULTS:

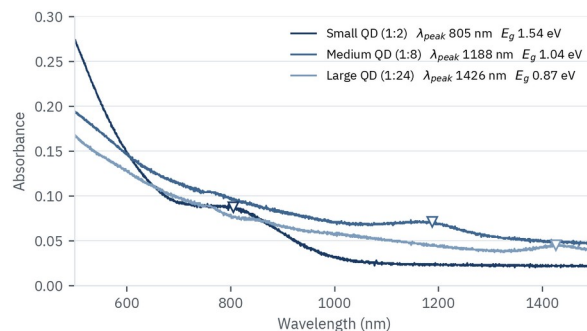


FIG 3: Absorbance of the three as-synthesized dot sizes, drop-cast on glass. Peaks at 805, 1188 and 1426 nm give band gaps of 1.54, 1.04 and 0.87 eV. The broad, smooth peaks indicate wide size dispersion.

Quantum dots. The three syntheses landed where intended. Absorption peaks at 805, 1188 and 1426 nm correspond to band gaps of 1.54, 1.04 and 0.87 eV, confirming that the oleic-acid ratio alone set the absorption edge across the full near-infrared. The peaks are broad rather than sharp, which is the first warning sign in the data: a monodisperse population gives a narrow excitonic peak, and these do not.

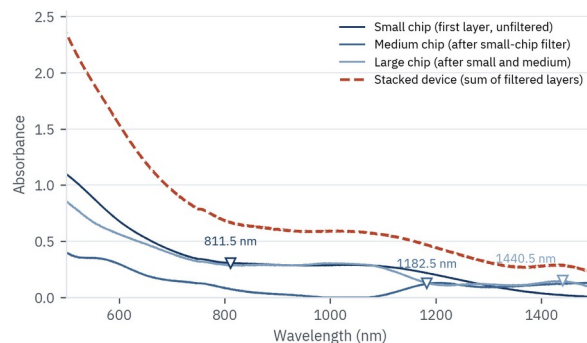


FIG 4: Filtered absorbance. Each curve shows what a layer actually sees after the layers in front of it have absorbed their share. The stacked device is the sum, spanning 500 to 1500 nm with peaks at 1182.5 and 1440.5 nm.

How the stack filters light. This is the mechanism the architecture depends on, and the absorbance data shows it directly. Light reaches the smallest dots first. Those dots have the widest band gap, 1.54 eV, so they absorb every photon carrying at least that much energy, which means everything from 500 nm out to roughly 1100 nm. Photons of lower energy cannot excite an electron across that gap, so they pass through the first chip untouched and arrive at the medium layer, whose 1.05 eV gap catches the 1100 to 1300 nm band. Whatever survives both reaches the large dots at 0.86 eV, covering 1300 to 1500 nm. Each layer is therefore both a detector and a high-pass filter for the one behind it, and the device as a whole spans

500 to 1500 nm without any layer competing with another for the same photons.

The filtering shows up in the photocurrent. The optical argument is confirmed electrically. When the medium chip is measured alone it sees the full spectrum, but placed behind the small chip its photocurrent falls by roughly half, because the high-energy photons that drove most of its signal have already been absorbed. The large chip loses about 80% of its photocurrent in the stack for the same reason, with two layers ahead of it. That drop is not a fault. It is the filter working, and it is the reason the stacked device covers a wider band than any single layer can.

Parameter	Gen 1 (Au / Si)	Gen 2 (ITO / glass)	Commercial Si
Dark current	415 μ A	9.3 μ A	~10 nA
Photocurrent	9.8 μ A	136 nA	~500 μ A
EQE	1.60%	0.022%	~77%
Responsivity	8.2 mA/W	114 μ A/W	~0.60 A/W
Rise / fall	0.46 s / 2.3 s	1.4 s / 4.1 s	~300 ns / 400 ns

TABLE 1: Figures of merit at 635 nm and 1.2 mW, against a typical commercial silicon photodiode.

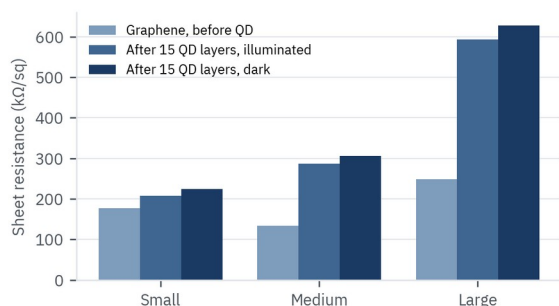


FIG 5: Graphene sheet resistance before QD deposition, and after fifteen layers under illumination and in the dark. TBAI treatment and darkness both raise resistance.

Electrical behaviour of the channel. Current flows through three possible paths, glass, graphene and the QD film, but graphene dominates because the other two are orders of magnitude more resistive. Sheet resistance rose after QD deposition on every chip, by 1.39 times on the small chip and 1.15 times on the medium, which is attributed to TBAI's flexible carbon chain interacting with graphene's π system and reducing carrier mobility. A second, smaller rise of 6 to 8% appears in the dark on all three chips, consistent with ambient photons exciting carriers in the graphene itself.

Time response and what it reveals. Rise time reports how quickly carriers generated in the dots reach the channel, and so reports dot quality; fall time reports how long excited carriers persist, and so reports contamination. The small and medium chips give roughly square responses, while the large chip is visibly smoothed and rises in 4.15 s, marking its dots as the

poorest of the three. Fall times are similar across all three, suggesting a common and uniform level of organic contamination rather than a problem with any one synthesis.

Where the efficiency went. Five causes account for the gap to commercial parts. The dots were a year old, and the broad absorption peaks in Figure 3 confirm size dispersion and organic contamination. Fifteen layers per chip, against two in the first generation, lengthened carrier transit and tripled rise time. ITO's conductivity is three orders below gold's and the graphene-ITO interface presents an energy barrier. TBAI raised sheet resistance 15 to 40% on the small and medium chips. Finally the largest chip showed edge patterning and poor QD coverage under a spin recipe developed for smaller substrates, explaining its 80% photocurrent loss independently of the filtering effect.

APPLICATIONS:

A detector whose spectral response is programmed at synthesis, coated onto glass at room temperature and read out through a transparent electrode is inherently scalable and inexpensive, with an active-material cost of order \$2 per m² against \$150 to \$165 per unit for commercial InGaAs and germanium photodiodes. With fresh, monodisperse dots and an engineered contact stack it suits applications where signal power is adequate and cost dominates: free-space optical links, industrial process imaging, agricultural and food-quality sensing, and low-cost NIR spectroscopy. The transparent substrate also opens uses closed to silicon and InGaAs, including window-integrated sensing, stacked multi-band imaging and large-area detectors.

The transferable result is the characterization strategy. Splitting a stacked device into independently measurable layers costs some performance, and here it cost photocurrent, but it converts an opaque result into a diagnosable one. Every number in this paper can be attributed to a specific layer, interface or process step, which is what makes the next iteration a matter of engineering rather than guesswork.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com